HAPPIEST HEALTH | DIGITAL TWIN CHALLENGE 2026

PulseTwin

A context-aware diabetes digital twin PoC

Demonstrating a two-hour synthetic glucose-excursion trend projection by pairing a simulated wearable stream with historical EHR context.

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SYNTHETIC RESEARCH DEMO - NOT A MEDICAL DEVICE

Make one risk concrete

A focused, testable prototype is more credible than claiming a whole-body digital twin.

Condition

Type 2 diabetes in a synthetic patient context, with historical HbA1c, BMI, diagnoses, and medication.

Demonstration target

Flag when the recent synthetic glucose trend extrapolates across an illustrative 180 mg/dL threshold within a 120-minute horizon.

- Intended interaction: a clinician-facing concept dashboard for inspecting inputs, trend direction, and model rationale.

Two streams, one inspectable twin

Every input and output in this prototype is synthetic and labelled as such.

Static EHR

Synthetic profile: demographics, diagnoses, BMI, HbA1c, medications.

FastAPI

Patient profile endpoint, accelerated WebSocket stream, scenario controls.

Demo models + UI

Current-input index, two-hour trend estimate, component explanation.

Simulated sensor stream

Glucose + heart rate; each one-second tick represents five simulated minutes.

Privacy-first by design

The public PoC contains no real identifiable patient data.

Historical / static

Local synthetic EHR JSON
Demographics and diagnoses
HbA1c and BMI
Medication list

Dynamic / simulated

Generated glucose (mg/dL)
Generated heart rate (bpm)
Five-minute virtual intervals
Seeded counterfactual traces

- No external or open clinical dataset is used; no model training or clinical validation is claimed.

Separate current state from future trend

Transparent demonstration logic; no trained machine-learning model or outcome labels.

Current-input index

Weighted demo score from current glucose, short-term change, HbA1c, BMI, and heart rate. Components are shown; score is not a probability.

Two-hour extrapolation

Least-squares line over up to 12 five-minute samples. Report projected glucose, slope, crossing time, and threshold state. Requires 3 samples.

- The projected contextual index combines extrapolated glucose with synthetic EHR HbA1c/BMI and current heart rate.

What the dashboard makes visible

A concept UI for exploring an evolving synthetic virtual patient.

Live overview

Synthetic patient banner
Glucose + heart-rate readings
Current-input index and components

Forecast view

Historical glucose curve
Dashed two-hour projection
Illustrative crossing status

Explore and export

Guided scenario replay
Same-profile counterfactual
JSON and PDF snapshots

- All controls and reports retain synthetic-data and non-clinical-use notices.

What the scripted traces show

Software behavior only - not model accuracy or a clinical outcome.

Steady synthetic trace

The seeded same-profile trace stays below the illustrative crossing condition in the demonstrated comparison.

Rising synthetic trace

The comparison flags a projected crossing during the scripted rise; later samples are already above the demo threshold.

- Counterfactual control: same EHR profile and heart-rate sequence; glucose trajectory changes.

A prototype is not clinical evidence

We are explicit about what this model cannot establish.

- Synthetic data only; the excursion threshold is an illustrative software setting, not a clinical cutoff.
- Linear extrapolation assumes the recent slope continues; it can overshoot when a trend turns or plateaus.
- No outcome labels, external benchmark, sensitivity, specificity, calibration, or prospective evaluation.
- Neither output supports diagnosis, medication changes, treatment decisions, or patient triage.

Not for diagnosis, treatment, or medication decisions.

How we would evaluate responsibly

Move from a behavior demo to a measurable research prototype.

- Select a documented, appropriately licensed synthetic/open dataset with event labels.
- Predefine the glucose-excursion target, prediction horizon, and evaluation cohort.
- Split by patient; prevent temporal and patient-level leakage.
- Compare to persistence and simple trend baselines; report event sensitivity, false alerts, lead time, and calibration.
- Review definitions and interpretation with clinical and data-governance experts.

A clear 20-minute walkthrough

Suggested recording outline; the team still needs to record and link the video.

Video run of show

Problem + use case: 3 min
Data + privacy: 3 min
Architecture + model: 4 min
Live walkthrough: 7 min
Limitations + next steps: 3 min

Before submission

Add the recorded video URL
Confirm public GitHub repository
Verify team registration details
Confirm each member approves public details

- Open-source license: MIT (see repository LICENSE).